

Null Space

The null space of a matrix A is the set of all vectors x such that $Ax=0$. In other words, it is the set of all solutions to the homogenous equation: $Ax=0$.

Example:

$$Ax = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\begin{matrix} 4 \times 3 & 3 \times 1 & 4 \times 1 \end{matrix}$
 $\hookrightarrow C(N)$ in $\mathbb{R}^4$
 $\hookrightarrow$ in $\mathbb{R}^3$

Let us try finding the possible solutions of x :

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \\ -2 \end{bmatrix}, \dots, \begin{bmatrix} c \\ c \\ c \\ -c \end{bmatrix}$$

$$\hookrightarrow c \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

Here, matrix A represents a linear transformation.

The null space x represents the set of all vectors that are sent to the origin by the matrix transformation. These vectors are flattened or collapsed in such a way that their direction and magnitude are lost, leaving only the zero vector.

How about $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$

$\hookrightarrow$ Does this give a subspace?

If not, why?

At first let us try finding some solutions:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix} \text{ and so on.}$$

but the general solution looks like: $\begin{bmatrix} 1-t \\ -t \\ t \end{bmatrix}$

(we will solve this later on)

⊛ Aside being closed under vector addition and scalar multiplication, the subspace must also contain the zero vector.

The general solution $x = \begin{bmatrix} 1-t \\ -t \\ t \end{bmatrix}$ does not contain any zero

The general solution $x = \begin{bmatrix} 1-t \\ -t \\ t \end{bmatrix}$ does not contain any zero vector. The solution set x of $Ax=b$ is not a subspace.

Solving $Ax=0$.

Let us take an example

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} \text{pivot} & & & \\ \text{columns} & \text{free} & \text{pivot} & \text{free} \\ & \text{columns} & \text{columns} & \text{columns} \end{bmatrix} = U \rightarrow \text{REF}$$

$\hookrightarrow$ rank of $A = \#$ of pivots $= 2$

⊛ You can assign any values to free columns to find the value of pivot columns

From A , $x_1 + 2x_2 + 2x_3 + 2x_4 = 0$

$$2x_3 + 4x_4 = 0$$

$$2x_3 + 4x_4 \xrightarrow{x_4=0} x_3 = 0$$

$$\hookrightarrow x_1 + 2x_2 = 0 \xrightarrow{x_2=1} x_1 + 2 \cdot (1) = 0 \rightarrow x_1 = -2$$

$$\therefore x = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \text{ and, } x = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

But that is not the whole null space. We can choose more combinations of free variables.

$$2x_3 + 4x_4 \xrightarrow{x_4=1} 2x_3 = -4 \rightarrow x_3 = -2$$

$$x_1 + 2x_2 + 2x_3 + 2x_4 = 0 \xrightarrow{x_2=0} x_1 + 0 - 4 + 2 = 0 \rightarrow x_1 = 2$$

So, $x = d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$, also a null space.

$$\therefore x = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

↳ a linear combination of all special solutions. (What is a special solution?)

↳ The special solutions correspond to specific choices of the free variables (x_2 and x_4 in this case) that is set to certain values that gives us the structure of the null space of the matrix, which is the set of all solutions to $Ax = 0$.

↳ The special solutions form the basis for the null space of the matrix, and they reveal the dependency structure between the variables in the system.

⊗ For a matrix m rows, n columns, n variables with rank r , meaning r pivot variables, we will have $n-r$ free variables.

Here, the REF of $A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Here, $\left. \begin{array}{l} Ax = 0 \\ 0x = 0 \\ 0x = 0 \end{array} \right\}$ are all same

↳ RREF of $A = \begin{bmatrix} \boxed{1} & 2 & 0 & -2 \\ 0 & 0 & \boxed{1} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$

↳	the pivot part	the free part
	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix}$
	$\mathbb{I}$	$\mathbb{F}$

typical RREF:

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \rightarrow \pi \text{ pivot } \pi_{\text{ous}}$$

$\uparrow$ $\uparrow$
 π pivot $n-n$
 columns free columns

If $N = \text{nullspace matrix}$

$$RN = 0, \quad N = \begin{bmatrix} -F \\ I \end{bmatrix}$$

$$R_x = 0$$

$$\begin{bmatrix} \mathbf{P} \\ \mathbf{F} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{pivot} \\ \mathbf{x}_{free} \end{bmatrix} = \mathbf{0}$$

Hence, $x_{\text{pivot}} = -\bar{F} x_{\text{free}}$



Computing the Nullspace by Elimination: $A = CR$

Before we compute nullspace, let us learn how to find $A=CR$ in a proper manner.

Example: $A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & 2 \\ 3 & 7 & 6 \end{bmatrix}$

(REF) $\rightarrow$

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 0 & -2 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

$\uparrow \qquad \qquad \uparrow \qquad \qquad \uparrow$
pivot pivot non
col col pivot

1. Find REF to identify pivot columns.
2. Form Matrix C with the pivot columns.
3. Form Matrix F with the non-pivot columns as linear combinations

$$\therefore C = \begin{bmatrix} 1 & 3 \\ 2 & 4 \\ 3 & 5 \end{bmatrix}$$

Here, third column is the linear combination of first and second column.

$$\text{So, } \alpha \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 6 \end{pmatrix}$$

$$\hookrightarrow \alpha = -5, \beta = 3.$$

$$\therefore R = \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

(Not the efficient method, look at the end for the efficient one)

A deeper look into $R = [I \ F]$

R is a matrix that is RREF of A removing the zero rows.

R_0 is a matrix that is RREF of A including the zero rows.

Example:

Simplest case

$$R = [I \ F]$$

$$r=2$$

$$m=2$$

$$n=4$$

$$\begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 4 & 6 \end{bmatrix}$$

General case

$$R_0 = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} P$$

$$r=2$$

$$m=3$$

$$n=4$$

$$\begin{bmatrix} 1 & 7 & 0 & 8 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\hookrightarrow$ I and F have r rows [$r \rightarrow \text{rank}$]

$\rightarrow$ Reduced matrix R_0 and the original have m rows.

$\rightarrow$ R_0 has $m-r$ rows of zeroes

Let us take a look at another example

$$A = \begin{bmatrix} 1 & 2 & 11 & 17 \\ 3 & 7 & 37 & 57 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 7 & 37 & 57 \end{bmatrix}$$

$\swarrow W$ $\searrow H$
 the leading 2×2 matrix

REF

$$A = \begin{bmatrix} 1 & 2 & 11 & 17 \\ 0 & 1 & 4 & 6 \end{bmatrix}$$

RREF

$$A = \begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 4 & 6 \end{bmatrix}$$

$\swarrow I$ $\searrow F$

After elimination, $W \rightarrow I$

$W \rightarrow$ independent columns

$H \rightarrow$ dependent columns

So, elimination = multiplying W^{-1} with A

$$\hookrightarrow W^{-1}A = W^{-1} \begin{bmatrix} W & H \end{bmatrix} = \begin{bmatrix} I & W^{-1}H \end{bmatrix} = \begin{bmatrix} I & F \end{bmatrix} = R$$

$$\therefore F = W^{-1}H$$

$$\Rightarrow WF = H$$

$$\Rightarrow H = WF$$

$\hookrightarrow$ The matrix F tells us how to combine the independent columns of A to produce the dependent columns

Another example:

$$A = \begin{bmatrix} 1 & 7 & 3 & 37 \\ 2 & 14 & 6 & 70 \\ 2 & 14 & 9 & 97 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 7 & 3 & 35 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 27 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 7 & 3 & 35 \\ 0 & 0 & 3 & 27 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 7 & 0 & 8 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here, $R_0 = \begin{bmatrix} 1 & 7 & 0 & 8 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\hookrightarrow R = \begin{bmatrix} 1 & 7 & 0 & 8 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Some key take away from this example are:

$\hookrightarrow R_0$ has a zero row because A has $m=3$ rows and $\text{rank}(A)=n=2$.
So $m-n=3-2=1$ row of zeros.

$\hookrightarrow I$ is in columns 1 and 3 of R_0 because those are the first independent columns of A .

$\hookrightarrow F$ in columns 2, 4 of R combines column 1, 3 of A to give its dependent columns 2, 4.

$\hookrightarrow H = \begin{bmatrix} 7 & 35 \\ 14 & 27 \\ 14 & 27 \end{bmatrix} = WF = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 0 & 9 \end{bmatrix}$

Now we will find special solutions of $Ax=0$ (using $A=CR$).

Example: $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 3 & 6 & 9 \end{bmatrix}$

$$\hookrightarrow R_0(A) = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow R = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and then } C = \begin{bmatrix} 1 & 1 \\ 2 & 5 \\ 3 & 9 \end{bmatrix}$$

This is pretty straightforward and better than the first one.

Let's re-visit that example again.

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & 2 \\ 3 & 7 & 6 \end{bmatrix}$$

$$\hookrightarrow \overset{REF}{A} = \begin{bmatrix} 1 & 3 & 4 \\ 0 & -2 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow A = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow R_0(A) = \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\hookrightarrow R = \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & 4 \\ 2 & 2 \\ 3 & 6 \end{bmatrix}$$

Now, revisit the previous example:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 4 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\hookrightarrow R_0(A) = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\hookrightarrow R = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \end{bmatrix}$$

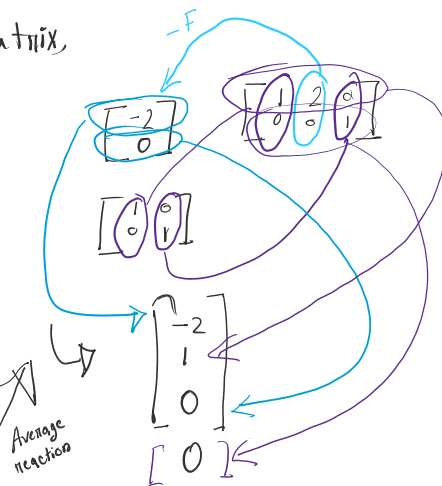
$$\hookrightarrow I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

Now, remember the cliff hanger at 3.2 Part 1?

If N is a nullspace matrix,

$$RN = 0 \rightarrow N = \begin{bmatrix} -F \\ I \end{bmatrix}$$

$$\Rightarrow N = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$



$\hookrightarrow -F$ is at first, so it is placed first.

$\hookrightarrow -F$'s first column is joint with first column of I .

$\hookrightarrow$ There is no second column of $-F$, so the second column of I cannot merged.

Key take-aways:

① For a π rank matrix with n columns, the number of free variables would be $n - \pi$.

$\hookrightarrow$ Also the number of

special solution.

- (ii) If a matrix has no dependent column, then there exists no free variables either. Thus the null space of that matrix is zero vector only.
- (iii) If a matrix has $n-r$ dependent columns, then the number of basis vectors of the null space are $n-r$. The dimension of the null space is $n-r$.